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Programme: MBA - Data Science and AI (3rd Semester)

Summer Internship (1st June - 12th July)

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Under the guidance of: Kuldeep Dubey

Internship Organization: Tutorsuite Technologies LLP, Kolkata

Academic Year: MBA 2023-25



CUSTOMER REVENUE INTELLIGENCE

An Interactive Data Science Platform for Customer Segmentation, Revenue Prediction and Business Growth

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Introduction

- Retail and e-commerce businesses record transaction data far faster than they convert it into decisions.
- Revenue is reported in aggregate - by month, category and region - but the questions that drive growth are customer-level: who matters most, who is going quiet, what will they spend next.
- Customer value is concentrated, so identifying and protecting high-value customers is more economical than acquiring new ones.
- Planning needs a forward view: predicted customer revenue and a forecast of company revenue with an honest error range.
- An integrated analytics platform is needed so segmentation, value analysis, prediction and forecasting all use one validated dataset and one definition of revenue.
- This project implements that platform on 18,670 validated order lines covering 1,984 customers over 36 months.

About the Organization

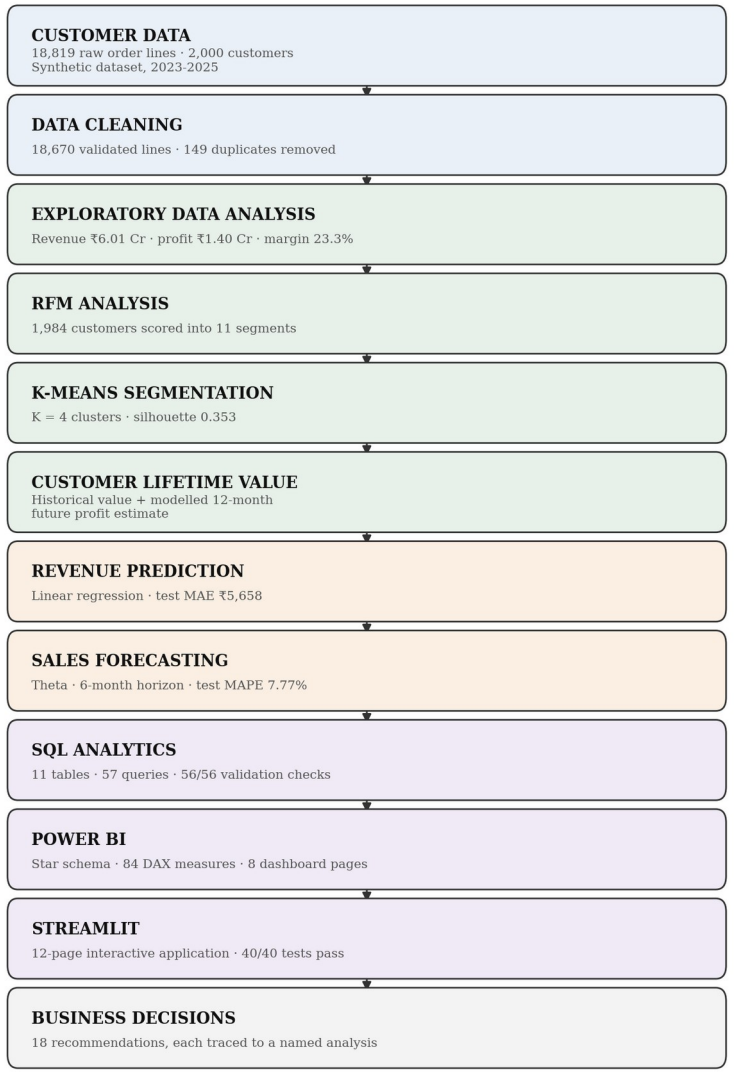
- **Internship organization: Tutorsuite Technologies LLP, Kolkata.**
- Internship period: 1st June - 12th July, under the guidance of Kuldeep Dubey.

- **Academic project based on a synthetic e-commerce transaction dataset.**
- The dataset was generated programmatically for academic demonstration. It does not describe any real company, customer or transaction, including the internship organization.
- It is reproducible from a fixed seed (42), which allows every result in this presentation to be regenerated and checked.
- No company history, revenue, client or operational information is presented, because none is documented for this project.

About Customer Revenue Intelligence

Customer revenue intelligence turns raw transactions into decisions by moving up a chain: who the customer is, what they are worth, and what they will do next.

Each stage consumes the validated output of the one before it, so one revenue definition holds end to end.



Review of Related Literature

- Six strands of published work inform the design. Full APA citations are on the References slide.

Area	Key work	Method	Relevance to this project
RFM analysis	Hughes (1994); Fader et al. (2005)	Quintile customer scoring	Rule-based customer value segments
Segmentation	MacQueen (1967); Rousseeuw (1987)	K-Means; silhouette	Unsupervised customer groups
Customer value	Gupta et al. (2006); Fader et al. (2005)	CLV modelling	Historical vs estimated forward value
Prediction	Hastie et al. (2009); Breiman (2001)	Regression and ensembles	Next-quarter customer revenue
Forecasting	Hyndman & Athanasopoulos (2021); Theta (2000)	ETS, ARIMA, Theta	Monthly revenue planning
BI delivery	Kimball & Ross (2013); Few (2006)	Star schema; dashboard design	Power BI model and application

Research Gap & Significance

- **The practical gap**
  - Transaction data -> traditional reporting -> limited customer intelligence -> limited predictive capability.
  - These techniques are well established individually, and prior research has combined several of them. What is rarely available in one reproducible workflow is all of them on one validated dataset with a single revenue definition.

- **The integrated solution implemented here**
  - RFM + K-Means + Customer value + Revenue prediction + Forecasting + SQL + Power BI + Streamlit.
  - Significance: every stage is re-runnable from a fixed seed, every metric is reconciled against an independent SQL implementation, and the whole project passes an audit of 135 checks (134 pass, 0 fail).

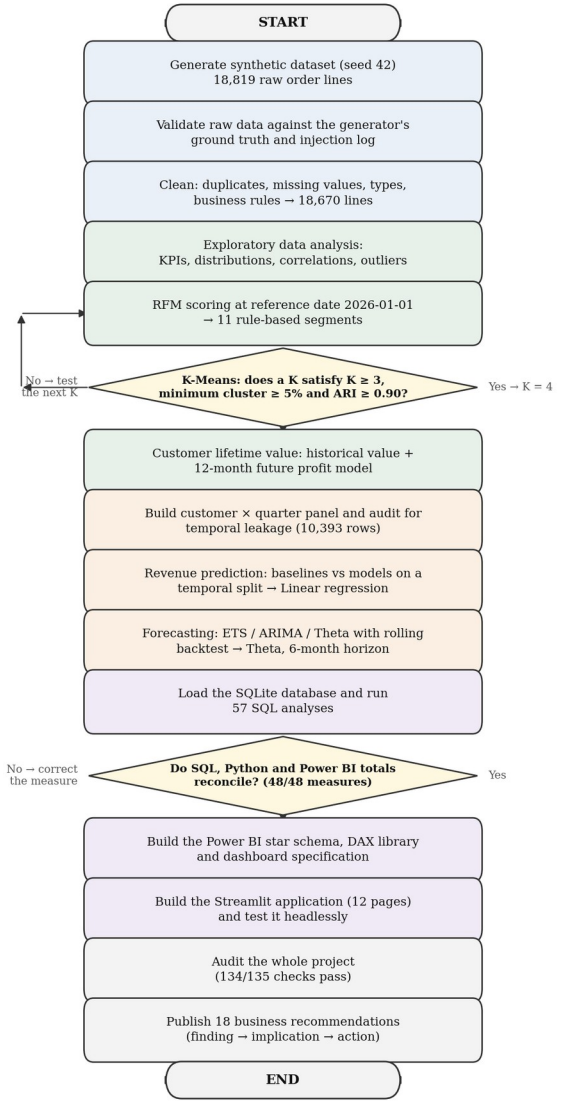
Objectives of the Study

- 1. Generate and validate the dataset, and clean it against explicit business rules
- 2. Characterise the revenue base through exploratory data analysis
- 3. Score every purchasing customer on Recency, Frequency and Monetary value
- 4. Segment customers independently using K-Means clustering
- 5. Quantify historical customer value and estimate forward twelve-month value
- 6. Predict customer revenue for the following quarter
- 7. Forecast company-level monthly revenue with prediction intervals
- 8. Reproduce the principal metrics independently in SQL and reconcile them
- 9. Build a Power BI semantic model and dashboard specification
- 10. Develop an interactive Streamlit application
- 11. Derive evidence-linked business recommendations

Research Methodology

Applied data science on quantitative transaction data: machine learning for segmentation and prediction, time-series analysis for forecasting, business intelligence for delivery.

Three validation gates let a stage reject its own output: business rules, K-selection and the forecasting baseline test.



Dataset & Data Preprocessing

- Raw data -> missing values -> duplicates -> data types -> business rules -> outliers -> clean dataset -> feature engineering.

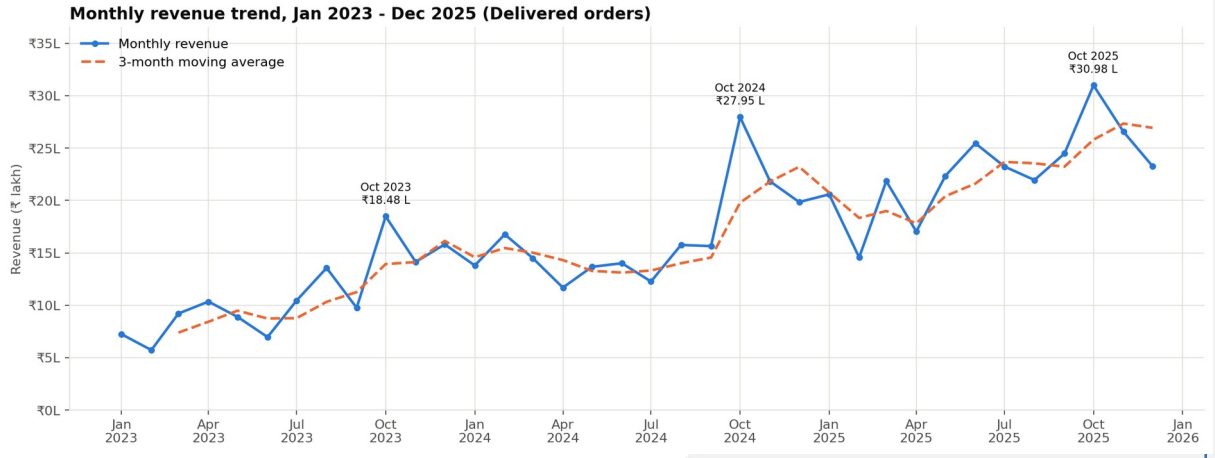
Dataset property	Value	Cleaning rule	Rows failing before / after
Raw extract	18,819 lines, 23 columns	Missing value	941 / 0
Cleaned dataset	18,670 lines, 51 columns	Invalid or non-ISO date	566 / 0
Duplicates removed	149	Non-numeric value	93 / 0
Customers	2,000 registered, 1,984 purchasing	Discount out of range	38 / 0
Products	61 in 6 categories	Revenue mismatch	56 / 0
Geography	35 cities, 21 states, 5 regions	Invalid category value	794 / 0
Period	01 January 2023 to 31 December 2025 (36 months)	Age out of range	19 / 0
Orders	13,251 across all statuses	Quantity not positive	9 / 0

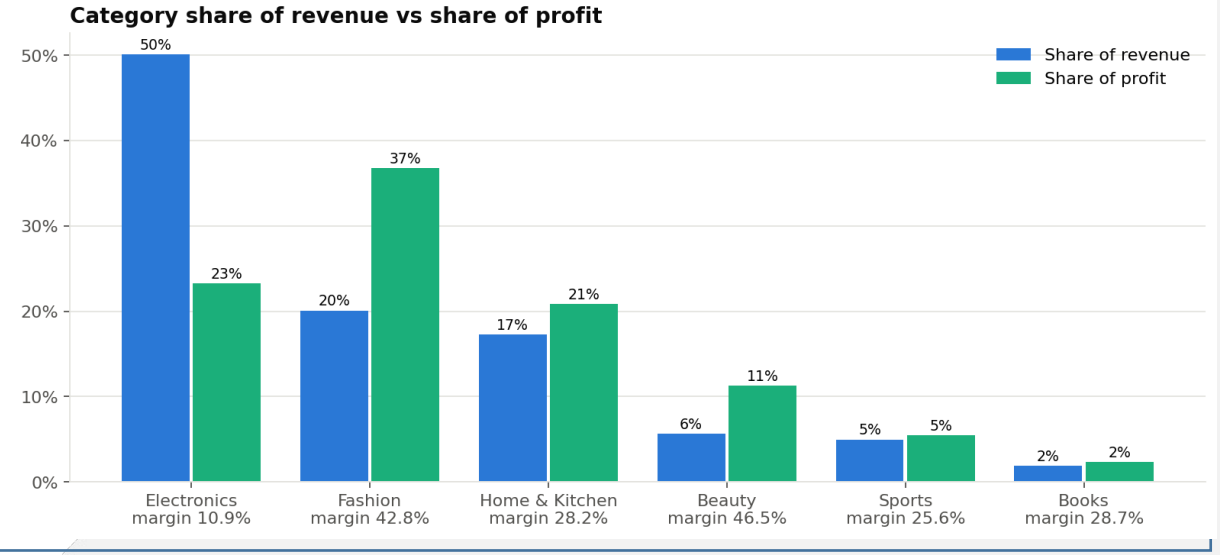
Exploratory Data Analysis

Delivered revenue Rs 6.01 Cr at 23.35% margin, 12,212 orders, AOV Rs 4,918.

Revenue grew 51.4% then 37.7% while margin rose to 23.90% - growth is not bought with margin.

The top 20% of customers produce 64.33% of revenue.



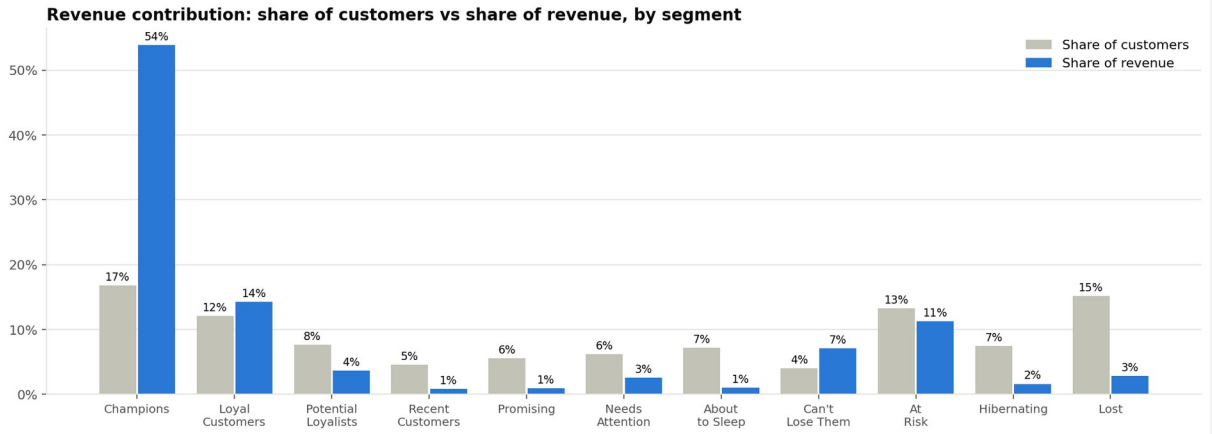


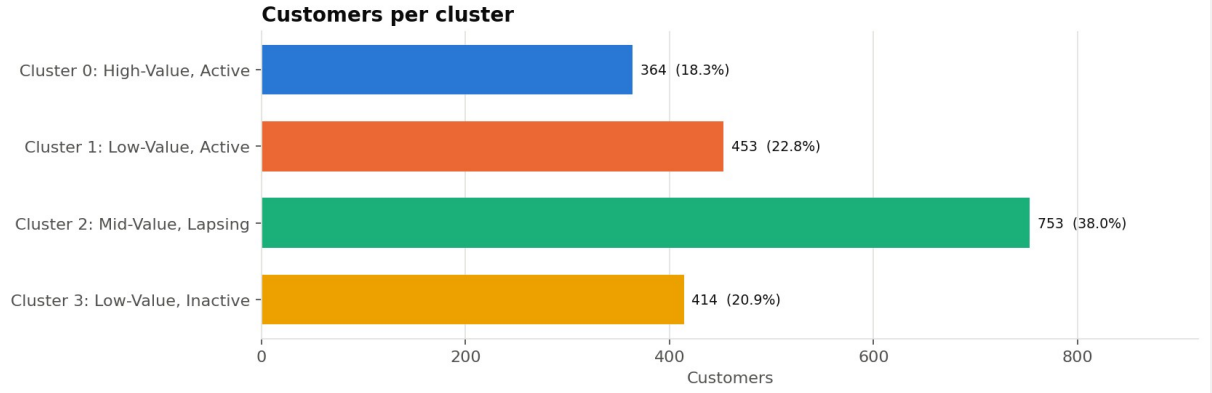
Customer Intelligence - RFM & K-Means

RFM scores 1,984 customers on recency, frequency and monetary value into 11 rule-based segments. Champions: 16.8% of customers, 53.8% of revenue.

K-Means on the same features selected K = 4 (silhouette 0.353). Cluster 0 is 18.3% of customers and 55.7% of revenue.

Agreement between the two schemes: ARI 0.3952.





Customer Lifetime Value

- Historical value (measured): Rs 1.40 Cr of delivered profit on Rs 6.01 Cr of revenue; mean Rs 7,068 against median Rs 3,893 per customer.
- Estimated forward value (predicted): a random forest trained out of time on eight behavioural features, predicting delivered profit over the next twelve months. The two are reported separately and are never added together.

Forward-value model	MAE	RMSE	R2	Spearman	Selected
Ridge regression (baseline)	Rs 2,853	Rs 4,469	0.2007	0.4779	
Random forest	Rs 2,793	Rs 4,269	0.2705	0.4869	Yes
Gradient boosting	Rs 2,982	Rs 4,323	0.2520	0.4960	
Naive: last-12-month profit repeats	Rs 2,947	Rs 4,648	0.1354	0.4402	
Simplified formula (AOV x F x margin x P(repeat))	Rs 2,859	Rs 4,606	0.1509	0.4641	
BG/NBD orders x avg profit per order	Rs 2,693	Rs 4,356	0.2405	0.4899	

Value drivers by permutation importance: Frequency, Avg_Discount, Recency_Days, AOV.

1,716 customers scored by the model, 268 by group average. 52 high-value customers are inactive for 180+ days, holding Rs 26.51 L of historical revenue.

Revenue Prediction

- Target: each customer's delivered revenue in the next quarter. Panel of 10,393 customer-quarter rows with 24 features computed strictly before the target quarter began.
- Out-of-time split (not random): 4,788 train / 2,692 validation / 2,913 test rows.

Model tested	Type	MAE	RMSE	R2	WAPE	Selected
Baseline: previous quarter (naive)	Baseline	Rs 6,423	Rs 14,483	-0.5700	145.1%	
Baseline: same quarter last year (seasonal naive)	Baseline	Rs 5,880	Rs 13,557	-0.3756	132.9%	
Baseline: 12-month quarterly mean	Baseline	Rs 5,436	Rs 11,466	0.0160	122.8%	
Linear regression	ML model	Rs 5,658	Rs 11,232	0.0557	127.8%	Yes
Random forest	ML model	Rs 5,762	Rs 11,265	0.0502	130.2%	
Gradient boosting	ML model	Rs 5,845	Rs 11,383	0.0302	132.1%	
Random forest (tuned)	ML model	Rs 5,554	Rs 11,148	0.0698	125.5%	

Selected: Linear regression by the rule fixed in advance - simplest model within 2% of the best validation RMSE (fixed in advance).

Most important features (permutation): Orders_12M, Main_Category, Share_Books, AOV_12M.

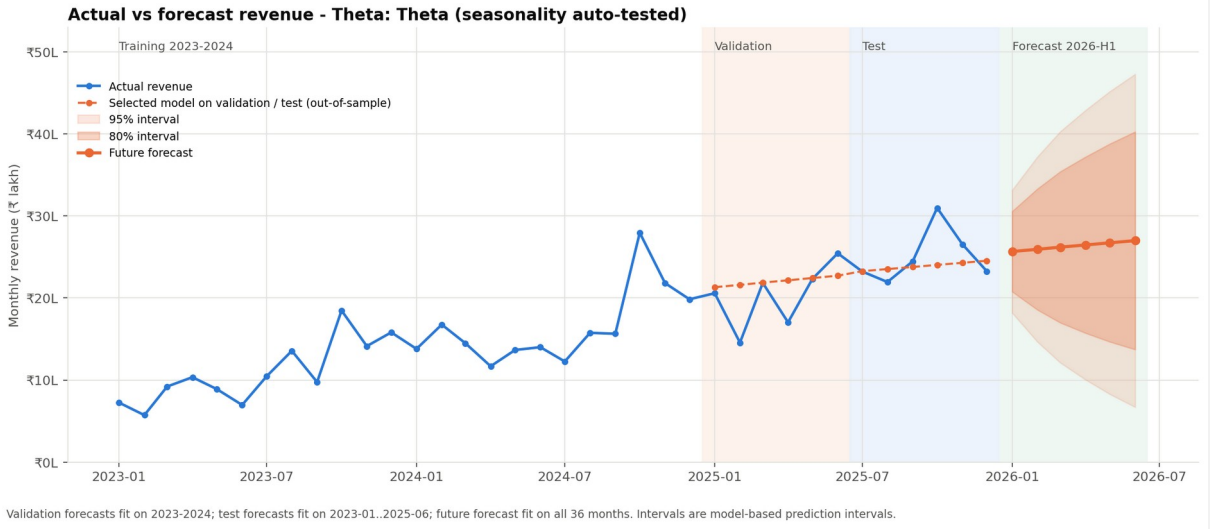
Honest reading: the model beats every baseline but is weak in absolute terms (R2 0.0557), because 53.0% of customer-quarters have a zero actual value. Total bias 1.12%, so it is usable at portfolio level, not for one customer.

Sales Forecasting

Forecasting is company-level monthly revenue; prediction is per customer per quarter. They are different tasks.

Naive, seasonal naive, ETS, ARIMA/SARIMA and Theta were tested on 36 months. Selected: Theta: Theta (seasonality auto-tested), backtest mean MAPE 14.56%.

6-month horizon: Rs 1.58 Cr, 80% interval Rs 1.00 Cr to Rs 2.16 Cr.



SQL-Based Business Analytics

- A normalised SQLite database of 11 tables and 2 views, with 57 analytical queries (37 analysis, 20 business). All 56/56 validation checks and 20 data-quality checks pass.
- Techniques used: JOIN, GROUP BY, common table expressions, window functions, self-joins, conditional aggregation and ranking.

Business question answered in SQL	Result computed	Source query
Revenue	Delivered revenue ₹6.01 Cr, profit ₹1.40 Cr (23.35% margin), 12,212 orders, 1,984 customers, AOV ₹4,918.	rev_kpi_summary
Growth	Revenue growth 2024: +51.4%, 2025: +37.7%; margin 22.65% -> 23.05% -> 23.90%.	rev_by_year
Customer concentration	Top 20% of customers = 64.33% of revenue; top 1% = 9.05%.	cust_concentration_summary
Repeat purchasing	Repeat customers are 89.4% of customers and 98.6% of revenue.	BQ12
Retention	Retained / active last year: 2024: 73.0%, 2025: 93.3%; new customers per year 2023: 1,137, 2024: 498, 2025: 349.	BQ15
RFM segments	Champions: 16.8% of customers, 53.8% of revenue; At Risk + Can't Lose Them: 18.4% of revenue.	seg_rfm_summary

This layer is an independent implementation: written against the same definitions in a different language, so agreement with the Python pipeline is evidence rather than a tautology.

Power BI Executive Dashboard

Star schema: 13 tables, 9 relationships, 84 DAX measures in 10 folders, nine pages specified.

All 48/48 measure validations match SQL; all 32 consistency checks pass.

Power BI Desktop is Windows-only and was unavailable, so the page shown is a design mockup rendered from the real data - not a screenshot.

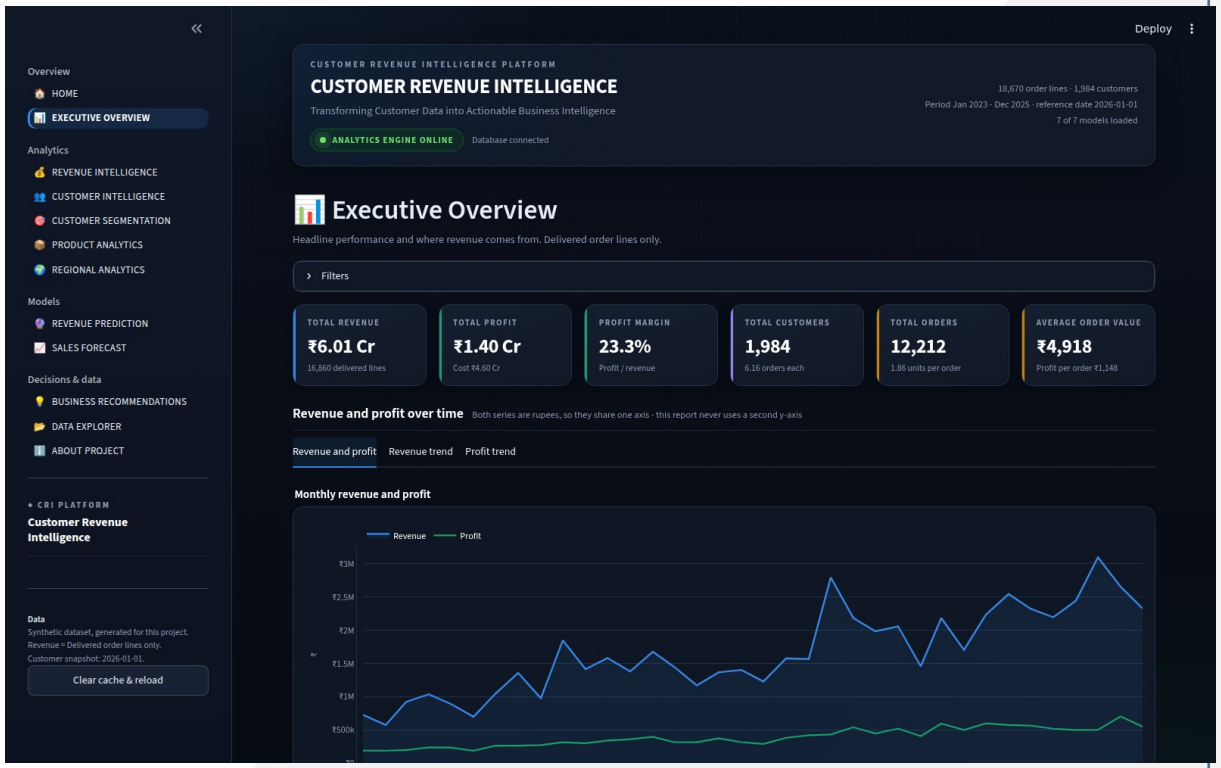


Interactive Streamlit Data Science Platform

A working 12-page application with interactive filters, Plotly charts, CSV download on every table and a data explorer.

It loads the persisted models rather than refitting, and computes no metric of its own - every number comes from the pipeline's outputs.

Tested headlessly: 40/40 checks pass. The image is a real screenshot.



Findings and Suggestions

- Key findings, and the recommendation each one supports:

Finding (from project output)	Business implication	Recommendation
Top 10% of customers produce 45.39% of revenue; top 20% produce 64.33%	Losing a few large customers costs more than a month of acquisition	Give the top decile a named owner and review it monthly
52 high-value customers inactive 180+ days, holding Rs 26.51 L of revenue	Value already earned is quietly at risk	Work the list as a contact campaign; record the outcome
Champions are 16.8% of customers but 53.8% of revenue	A small group carries the business	Protect and grow Champions before chasing new customers
Repeat customers are 89.4% of the base and 98.6% of revenue	The second order is where value is created	Target the first-to-second order conversion
Electronics is 50.10% of revenue at 10.85% margin; Beauty earns 46.49%	Volume leader and margin leader are different assets	Manage category mix rather than a single revenue target
Zero-discount band returns the best margin in 6 of 6 categories	Discounting is costing margin (association, not proven cause)	Run a controlled discount experiment before changing policy

Future scope of the study

- Validate the pipeline on a real transaction dataset under an appropriate data-protection agreement, and compare the segment structure that emerges with the one reported here.
- Define an operational inactivity label with the business, and convert the present recency flag into a supervised model that can be evaluated against it.
- Reframe next-quarter prediction as a two-stage problem - will the customer order at all, and if so how much - to address the zero-inflation that limits the present model.
- Extend the forecast with external regressors: promotion calendar, price index and holiday effects, benchmarked against the current univariate backtest.
- Run a controlled discount experiment to establish whether the observed discount-margin association is causal.
- Operationalise the pipeline with scheduled re-runs, drift monitoring on the input distributions and model-performance tracking against realised outcomes.
- Complete the Power BI build on a Windows machine from the supplied semantic model, measures and build guide.

Conclusion

- The project converts 18,670 validated transaction lines into an integrated decision-support platform, carrying one dataset and one revenue definition through cleaning, exploratory analysis, dual segmentation, customer value analysis, prediction, forecasting, SQL analytics and two delivery surfaces.
- Value is concentrated: the top 20% of customers hold 64.33% of Rs 6.01 Cr in delivered revenue, and two independent methods - RFM rules and K = 4 clustering - identify substantially the same high-value group.
- Reliability differs by horizon and is reported as such: monthly company revenue forecasts to a backtest MAPE of 14.56%, while individual customer revenue predicts weakly.
- The integration is tested, not asserted: an audit of 135 checks returns 134 passes and 0 failures across 39 Python modules, 4 SQL files and 9 executed notebooks.
- Every figure presented is read from the project's own output files, so the presentation, the report and the application cannot disagree.

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The complete reference list of 38 entries appears in the project report.

THANK YOU

Questions & Discussion

Swagata Das | MBA - Data Science and AI (3rd Semester)

CUSTOMER REVENUE INTELLIGENCE: An Interactive Data Science Platform for Customer Segmentation, Revenue Prediction and Business Growth